feature of the neural system is its ability to respond simultaneously to several stimuli, for instance, we can play piano and sing; listen to music and do household chores. In all such coordinated movements, whether skilled performances or routine tasks like cycling or driving, the integrating power of the neural system is involved. In this chapter, you will understand how neural system is organized; how it integrates all organs and what kind of cellular events underlie its functioning.

10.1 Neural System

The neural system comprises of highly specialized cells called **neurons**, which can detect, receive, process and transmit different kinds of stimuli. Simple form of neural system as nerve net is seen in lower invertebrates. The neural system of higher animals are well developed and performs the following basic functions:

- **Sensory functions**- It receives sensory input from internal and external environment.
- **Motor functions**- It transmits motor commands from the brain to the skeletal and muscular system.
- **Autonomic functions**- Reflex actions.

10.2 Human Neural System

The human neural system is divided into two**, the central neural system (CNS)** and **the peripheral neural system** (**PNS**). The structural and functional units of the neural system are neurons that transmit nerve impulses. The non-nervous special cells called **neuroglia** form the supporting cells of the nervous tissue.

There are three functional classes of neurons. They are the **afferent neurons** that take sensory impulses to the Central Neural system (CNS) from the sensory organs; the **efferent neurons** that carry motor impulses from the CNS to the effector organs; and **interneurons** that lie entirely within the CNS between the afferent and efferent neurons.

The central neural system lacks connective tissue, so the interneuron space is filled by neuroglia. They perform several functions such as providing nourishment to the surrounding neurons; involving the memory process; repairing the injured tissues due to their dividing and regenerating capacity; and acting as phagocyte cells to engulf the foreign particles at the time of any injury to the brain.

Glial cells do not lose the ability to undergo cell division; so most brain tumours of neural origin consists of glial cells. Neurons themselves do not form tumours because they are unable to divide and multiply.

10.3 Neuron as a Structural and Functional Unit of Neural System

A neuron is a microscopic structure composed of three major parts namely **cell body** (soma), **dendrites** and **axon.** The cell body is the spherical part of the neuron that contains all the cellular organelles as a typical cell (except centriole)**.** The plasma membrane covering the neuron is called **neurilemma** and the axon is **axolemma.** The repeatedly branched

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